

PHYS 230 - Electricity and Magnetism

Lecture 14

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

3 Electric Potential Energy and Potential

- Electric Potential (Continued)

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

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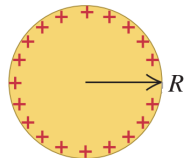
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Exercise I

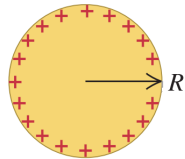
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A solid conducting sphere of radius R has a total charge q . Find the electric potential everywhere, both outside and inside the sphere.



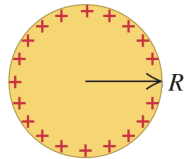
Exercise I

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Exercise I

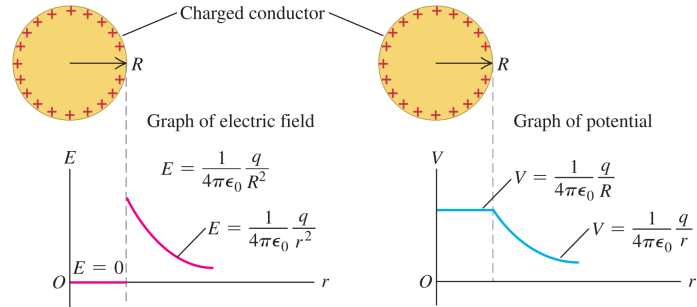
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Exercise 1

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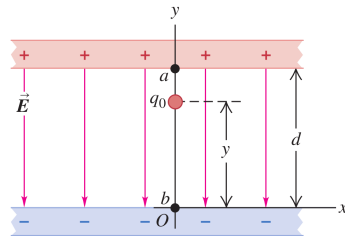
Figure 23.16 Electric-field magnitude E and potential V at points inside and outside a positively charged spherical conductor.



Exercise 2

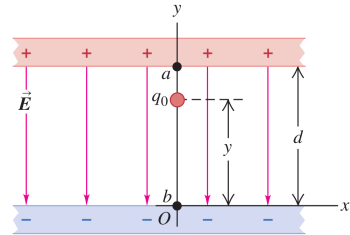
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- (a) What is the potential difference between the two parallel charged plates with uniform $\mathbf{E}$ between them going from a to b ?
- (b) Taking $U|_{y=0} = 0$, what is the potential of any point with a distance from the bottom plate?



Exercise 2

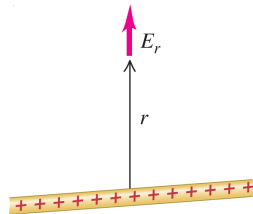
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Exercise 3

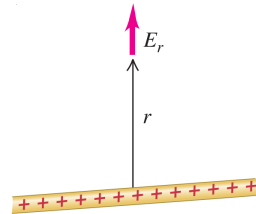
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Find the potential at a distance r from a very long line of charge with linear charge density λ .



Exercise 3

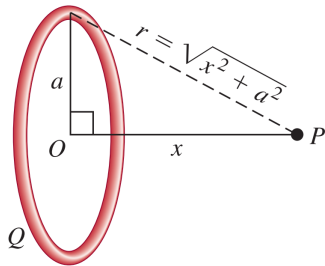
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Exercise 4

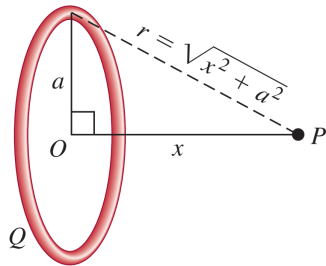
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Find the potential at a point P on the axis perpendicular to the plane of a ring of radius a with linear density λ and a total charge Q , at a distance x from the center of the ring.



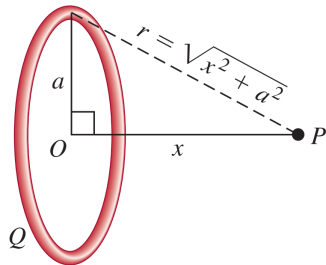
Exercise 4

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Exercise 4

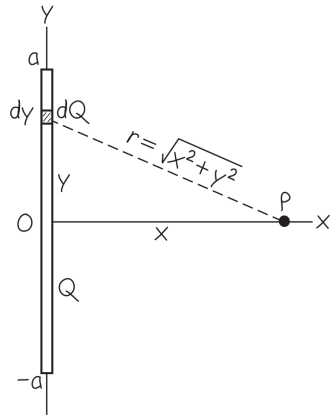
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Exercise 5

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Positive electric charge Q is distributed uniformly along a line of length $2a$ lying along the y -axis between $y = -a$ and $y = +a$. Find the electric potential at a point P on the x -axis at a distance x from the origin.



Exercise 5

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